

Major Advanced Topics

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Magnetic-Field Effect in High-Order Above-Threshold Ionization

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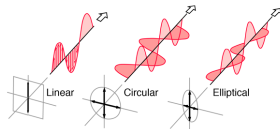
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Surrounding concepts

■ Polarization



■ Nondipole effects

$$H = \frac{1}{2} \left(\mathbf{p} + q\mathbf{A}(x, t) \right)^2 + V_0,$$

$$\mathbf{A}(x, t) \sim A(t) + \frac{x}{c} F(t) - \frac{1}{2} \frac{x^2}{c^2} \frac{d}{dt} F(t).$$

■ Relevant equations

$$\mathbf{F} = -\frac{\partial \mathbf{A}}{\partial t}, \quad \nabla \times \mathbf{B} = \frac{1}{c^2} \frac{\partial \mathbf{F}}{\partial t}.$$

■ Ponderomotive energy

$$U_p = \frac{F_0^2}{4\omega^2}.$$

■ Cross-section σ

Quantity proportional to the rate at which a particular radiation-target interaction occurs. If the incoming radiation is considered as being composed by quanta, then the cross section is a scattering rate.

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The state of the problem

■ Conservation of momentum

$$\langle p_{x,e} \rangle + \langle p_{x,\text{ion}} \rangle = \frac{E_\gamma}{c} = \frac{E_e}{c} + \frac{I_p}{c}.$$

■ How is the photon energy shared between electrons and ions?

- For high photon energies, $\langle p_{x,e} \rangle = 8/5 \times E_e/c$.
- For recollision free strong field ionization, $\langle p_{x,e} \rangle \sim E_e/c + 1/3 \times I_p$.
- For linearly polarized fields, around $2U_p$ region, $\langle p_{x,e} \rangle \sim E_e/c$.

■ For rescattered electrons whose energy can reach up to $10U_p$, the question is open. Still, Brennecke *et al.* predicted that for He, $\langle p_{x,e} \rangle$ levels off and remains constant.

■ Some interesting comments

- Requirements of high intensities and long wavelengths place HHG and LIED outside of the dipole oasis.
- Nondipole effects have been deepened by the analysis of NSI, field inhomogeneity, nonadiabaticity and weak coulomb effects...

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What to expect from this paper

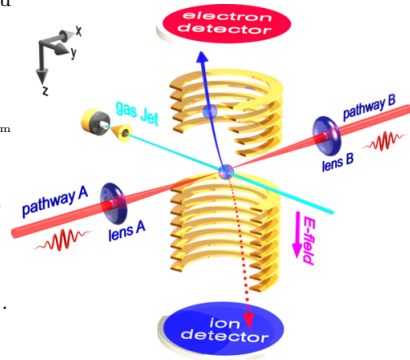
- First experimental results on the photon-momentum transfer for the ionization of Xe atoms by linearly polarized light beyond the classical cutoff.
- First experimental test of the Brennecke proposition, (which was about He).
- Quick experimental observations
 - A breakdown of $\langle p_{x,e} \rangle$ is found as soon as energy gets higher than $2U_p$.
 - Still, the plateau-like behaviour predicted by Brennecke for He is not seen.
 - Chances are that the momentum transfer is target sensitive.
 - Present paper results are for Xe.

Magnetic-Field Effect in High-Order Above-Threshold Ionization

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Experimental setup

- Linear-photon-momentum transfer $\langle \Delta p_x \rangle$ is around 4×10^{-4} a.u. for Ti:Sapphire (800 nm).
- Making use of the following strategy
 - (800 nm, 10 kHz, 25 fs), Coherent Legend Elite Duo
SANTA CLARA, Calif., July 1, 2010 – A new Ti:sapphire ultrafast laser amplifier from Coherent Inc., the Legend Elite Duo-HP, offers >15 W of output power and pulse energy of up to 10 mJ. The kilohertz-class thermoelectrically cooled amplifier can be configured to provide output pulse durations of <25 fs, <35 fs, <130 fs or 0.5 to 2 ps with repetition rates ranging from 1 to 10 kHz.
 - Split, then focused back from 2 opposite directions on the same spot of supersonic Xe gas.
 - In the reaction volume, 7×10^{13} W/cm² ($U_p = 4.2$ eV).
 - Electrons and ions guided to opposite directions with a static field of 29.8 V/cm.
 - Symmetry of the experiment along the polarization direction (z -axis) prompted the symmetrisation of the data in regard that same direction.

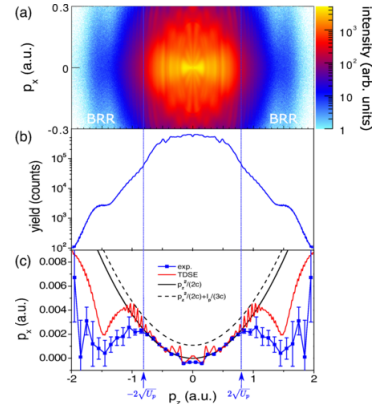


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Two-dimensional PEMD along the polarization z - and light propagation x - axis

- Figure 2(a) displays the 2D distribution of the particles momentum.
 - $|p_z| \leq 0.2$ a.u. electrons experience strong Coulomb focusing.
 - $0.2 \text{ a.u.} \leq |p_z| \leq 2\sqrt{U_p}$, distribution is dominated by photon holography and ATI rings are seen.
 - $2\sqrt{U_p} \leq |p_z|$, distribution dominated by HATI. Over the $10U_p$, BRR can be identified.
- Figure 2(b) shows the distribution (or the number) of electron having a precise momentum regarding the polarization direction.
- Figure 2(c) shows the distribution of transferred momentum p_x along the light propagation direction for each p_z .
 - For the intermediate energies, overall agreement.
 - For $p_z < 2\sqrt{U_p}$, experimental results shift negative values.
 - For $p_z > 2\sqrt{U_p}$, values deviate from the expected shape.



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Modelling and theoretical calculations

- Explanation of the local minimum structure.
- Simple backreflection model (BRM) based on some nondipole generalization of the 3 steps model.
 - First steps are the exact same.
 - Incident velocity is flipped around by the magnetic field and augmented by $v_{\text{in},x} = v_{\text{in},z}^2/(2c)$ along the light propagation direction.
 - For Xe^+ , the differential electron scattering cross-section has a local maximum at 180° (backscattering) which induces the maximum in the p_x -direction near the p_z -axis.
 - They therefore estimate the nondipole shift by "assuming" that all electrons are backscattered.
 - Recollision slows down the electron momentum along p_x -, while it is boosted along p_z -direction.
 - Two trajectories born per optical cycle contribute for the same p_z , termed "long-" and "short-".
- Low Frequency Approximation (LFA) is implemented in order to complete the above model.
 - Based on complex-valued electron trajectories.
 - Accurately incorporate the field free differential.
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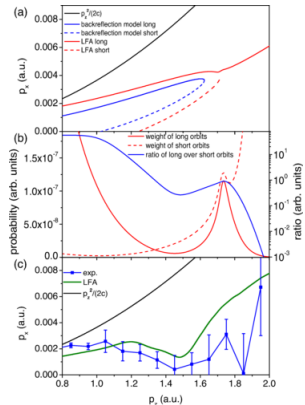
$$M(\mathbf{p}) = P(\mathbf{p})T[\mathbf{v}_{\text{in}}(\mathbf{p}), \mathbf{v}_{\text{out}}(\mathbf{p})], \quad \sigma \sim |T|^2.$$

Magnetic-Field Effect in High-Order Above-Threshold Ionization

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Result of the classical and quantum-orbit analysis

- (a) Momentum-shift estimate from classical BRM and LFA.
 - Nondipole shift in LFA long- and short- trajectories similar to the BRM.
 - Nonetheless, due to nonadiabatic initial velocities, difference between short- and long- orbits is larger.
- (b) Probability of quantum orbits with final momentum $p_x = 0$ in LFA.
 - Though both orbits contribute, the shift is strongly influenced by the relative weight between them.
 - Relative weight mainly determined by the energy-dependent cross-section.
 - Long- dominates for $p_z \leq 1.3$ a.u., as short- becomes comparable for $1.3 \text{ a.u.} \leq p_z \leq 1.7$ a.u.
 - Transition from long- to short-, then back to long-trajectory causing the minimum structure in the shift p_x .
- (c) Focal volume-averaged LFA result calculated for the Tong-Lin potential.
 - Reproduces the experimentally observed minimum structure.
 - Confirms the prediction for $p_z \geq 1.6$ a.u.

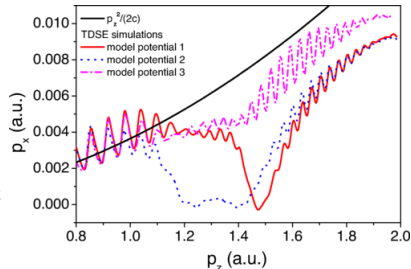


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Model potential dependence of the nondipole shift

- Explanation of the difference between He and Xe.
- Model potential $V(r)$ expression
 1. $-(Z + a_1 e^{-a_2 r} + a_3 e^{-a_4 r} + a_5 e^{-a_6 r})/r$ (Tong-Lin),
 2. $-2(N\Gamma - Z)/r$, with $\Gamma = 1 - [(e^{r/d})B + 1]$ (GSZ),
 3. $-(Z + a_1 e^{-a_2 r})/r$ (effective).
- Predicted shifts different in the HATI region, which implies that the electron-ion interaction plays an important role.
- Tong-Lin potential (1) gives the most accurate results.
- Comparison shows that shifts are very sensitive to the details of the scattering process.
- This also explains why results are different from He, since the specificities of the target are very important for the dynamics.
- Good agreement between theory and prediction for $p_z \leq 1.0$ a.u.



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Closure

- Partitioning of the linear photon momentum between electron and nucleus altered by the rescattering process.
 - Depending on the electron energy compared to $2U_p$, the share taken by the electron drastically different.
 - In the case of Xe, domination of the distribution by backscattered electrons in the $2\sqrt{U_p} \leq p_z$ region.
 - Recollision with the parent ion transfers most of the momentum to the core, which halves the momentum altogether, especially in the p_x direction.
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- Sensitivity of the nondipole shift to the inner structure of the parent ion turns it into a promising tool for atomic and molecular imaging.